Project Report Format

1. INTRODUCTION

Electric Vehicles (EVs) are becoming increasingly popular due to their environmental benefits and energy efficiency. With the growing number of EV models and charging stations, it is important to analyze and visualize EV-related data effectively. This project, Visualization Tool for Electric Vehicle Charge and Range Analysis, uses Tableau to present EV data through interactive dashboards and stories, making it easier to understand charging infrastructure, vehicle performance, and market trends.

1.1 Project Overview

The project focuses on analyzing electric vehicle data using Tableau. It provides interactive visualizations of charging stations, EV brands, prices, body styles, powertrain types, and vehicle performance. The dashboards and stories help users compare different EVs and gain meaningful insights from the data.

1.2 Purpose

The main purpose of this project is to develop an interactive visualization tool that simplifies EV data analysis. It helps users compare electric vehicles, understand charging infrastructure, and make informed decisions using clear and interactive dashboards.

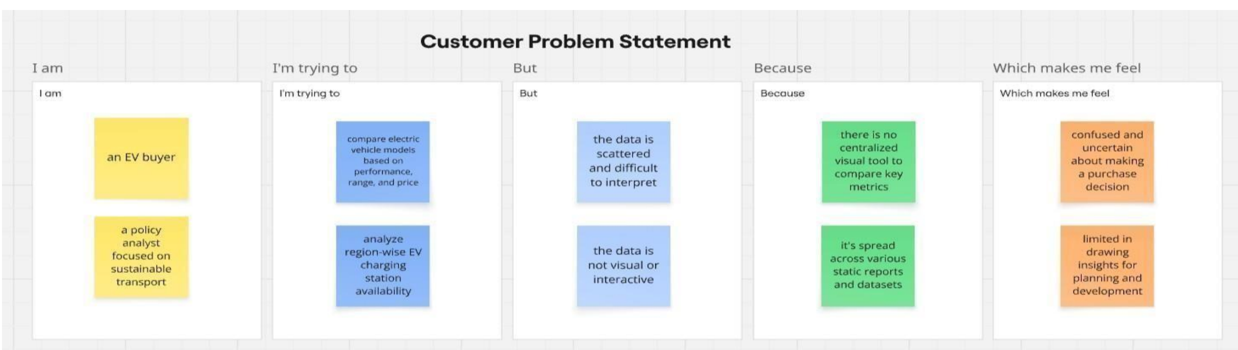
This version is concise, easy to understand, and suitable for your final report.

2.<u>IDEATION PHASE</u>

The ideation phase focuses on identifying the problem, understanding user needs, and developing a suitable solution. For this project, different ideas were discussed to analyze electric vehicle data effectively and present it using interactive dashboards and stories.

2.1 Problem Statement

The rapid growth of electric vehicles has resulted in a large amount of data related to charging stations, vehicle specifications, pricing, and performance. However, this information is often scattered across different sources, making it difficult for users to compare and analyze. This project addresses the problem by creating an interactive visualization tool that presents EV data in a simple, organized, and easy-to-understand format.



Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel

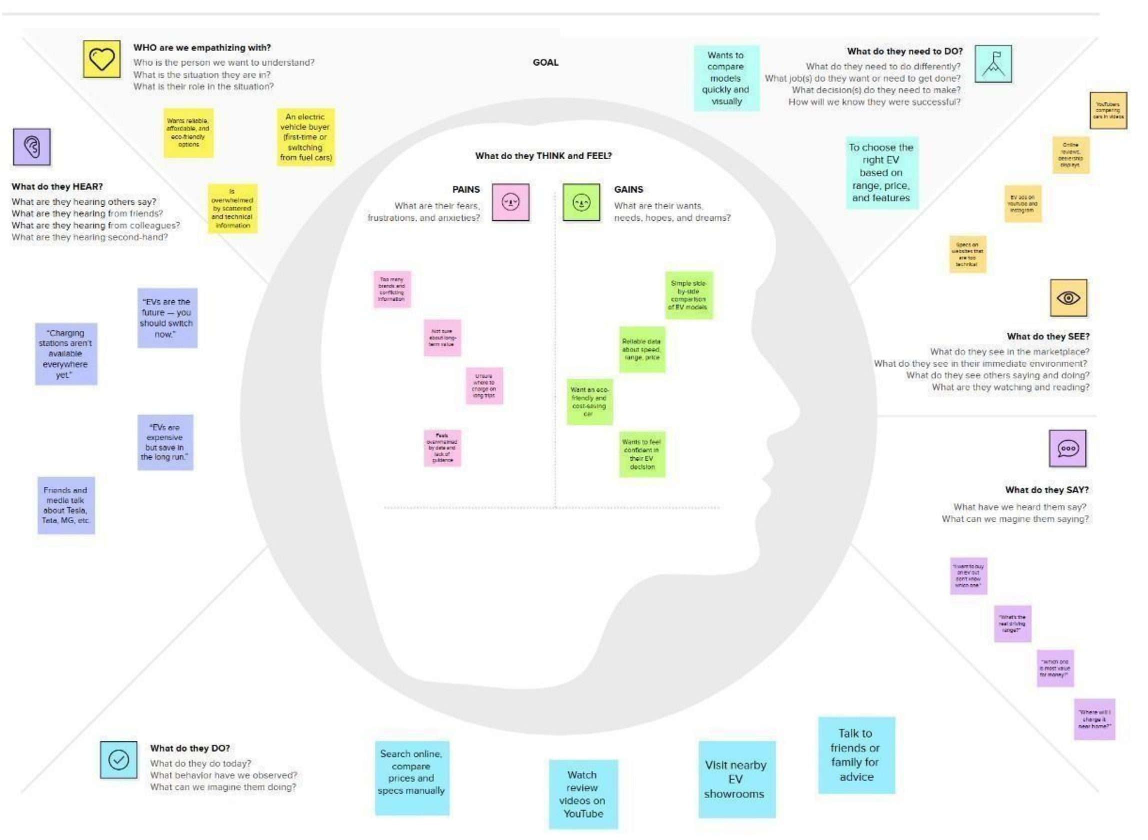
PS-1	an EV buyer	compare electric vehicle models based on performance, range, and price	the data is scattered and difficult to interpret	there is no centralized visual tool to compare key metrics	confused and uncertain about making a purchase decision
PS-2	a policy analyst focused on sustainable transport	analyze region-wise EV charging station availability	the data is not visual or interactive	it's spread across various static reports and datasets	limited in drawing insights for planning and development

2.2 Empathy Map Canvas

The empathy map was created to understand the needs and expectations of users interested in electric vehicles. It helps identify what users think, feel, see, and do while searching for EV information. Based on these insights, the dashboard was designed to provide clear visualizations and meaningful comparisons for better decision-making

An empathy map is a visual framework designed to organize and represent insights about a user's thoughts, emotions, actions, and experiences in a clear and structured way. It enables teams to develop a deeper understanding of user needs, expectations, and behaviors.

Developing an effective solution begins with identifying the actual problem and understanding the people affected by it. Creating an empathy map encourages team members to view the situation from the user's perspective, helping them recognize the user's objectives, motivations, pain points, and challenges. This process supports better decisionmaking and leads to the design of more user-focused and practical solutions.

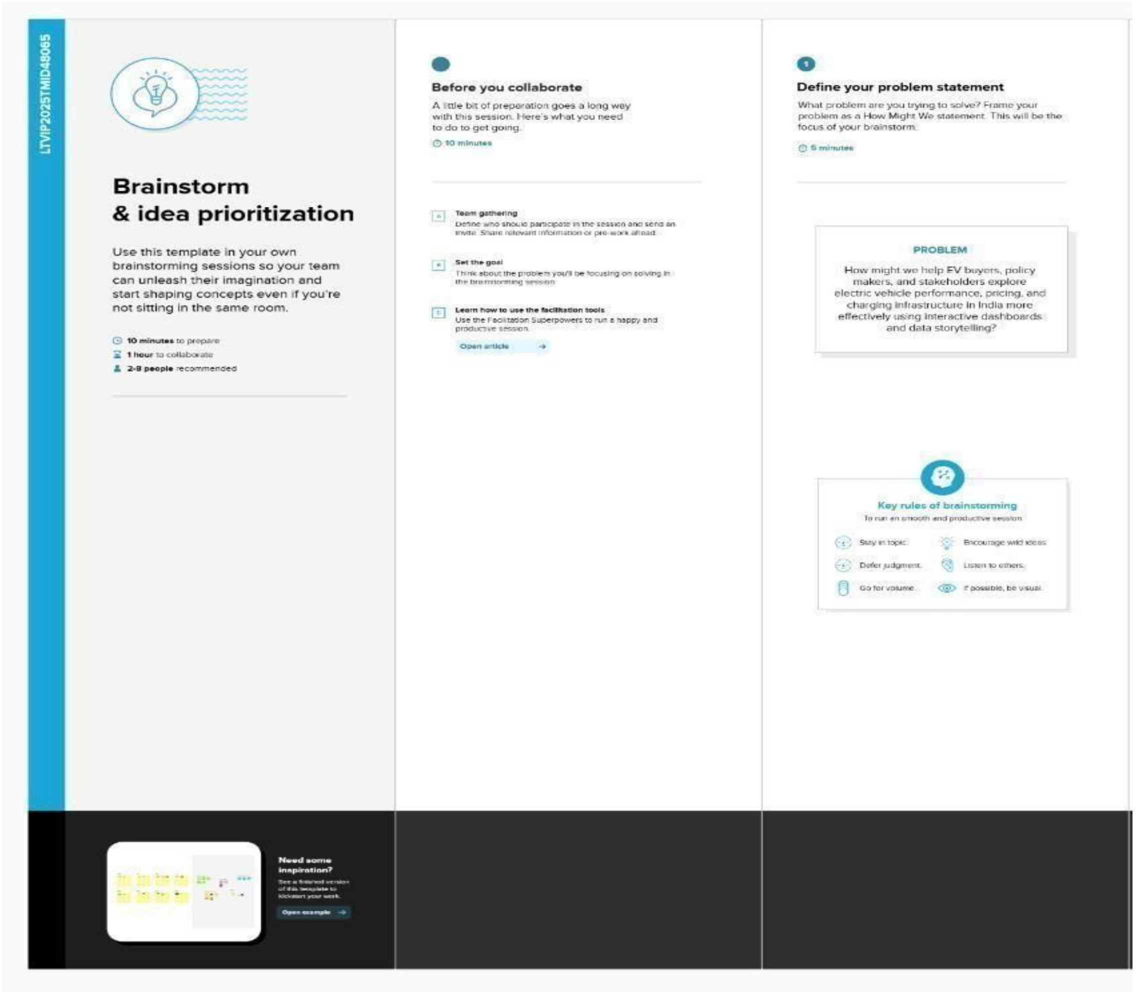


2.3 Brainstorming

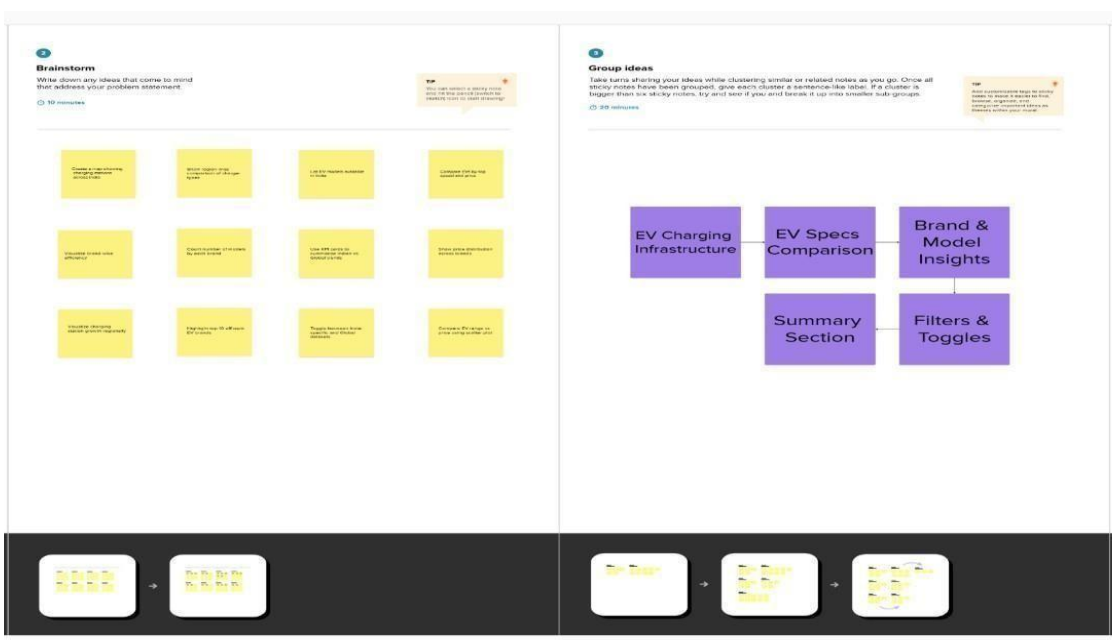
Brainstorming is a collaborative approach that allows team members to freely express and exchange ideas without immediate judgment. It encourages creativity by welcoming innovative and unconventional suggestions while promoting active participation from everyone involved. During this process, the emphasis is placed on generating a wide range of ideas rather than evaluating their quality at the beginning. These ideas are then discussed, refined, and organized collectively, enabling the team to identify practical and effective solutions for the selected problem.

During the brainstorming phase, different ideas were discussed to identify the most useful visualizations for EV analysis. The team decided to include dashboards showing charging stations, EV prices, vehicle efficiency, body styles, powertrain types, and brand comparisons. These visualizations help users explore EV data easily and gain valuable insights.

Step-1: Team Gathering, Collaboration and Select the Problem Statement



Step-2: Brainstorm, Idea Listing and Grouping

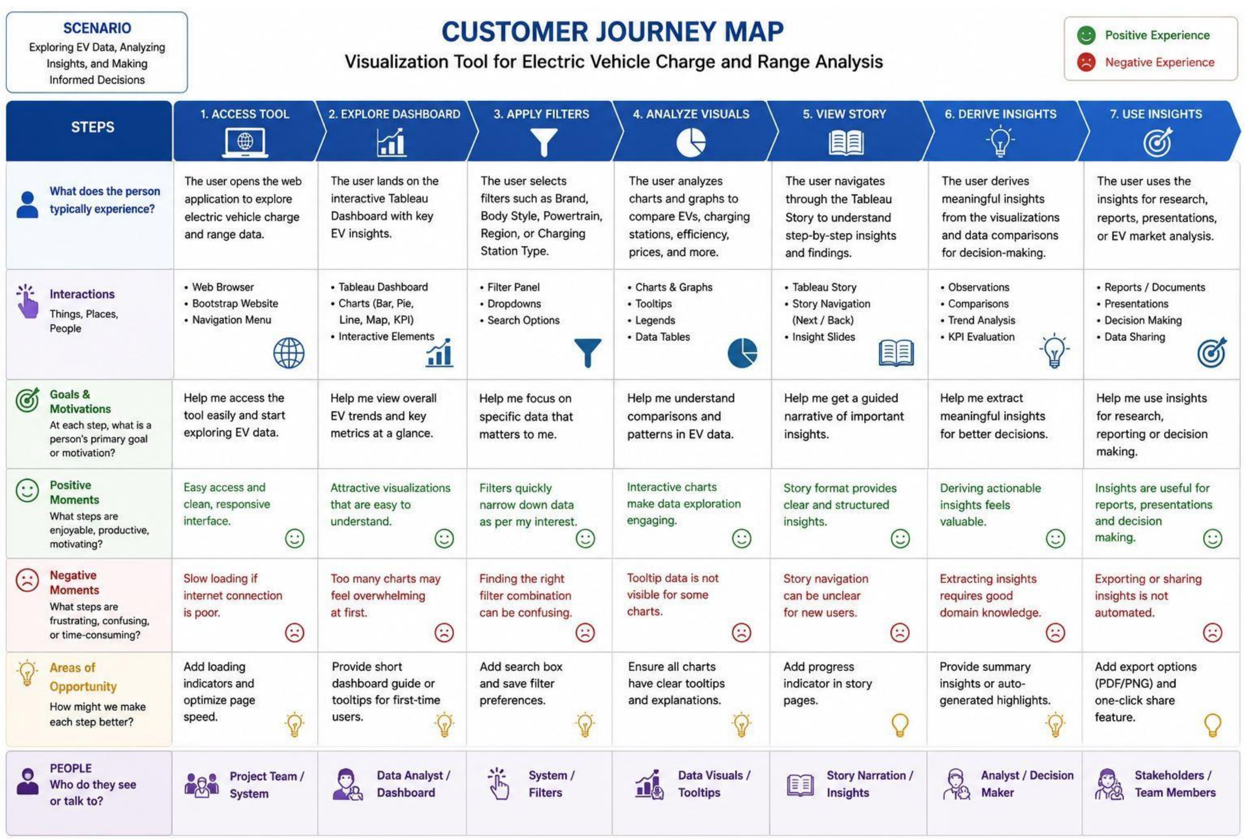


3.<u>REQUIREMENT ANALYSIS</u>

Requirement analysis is the process of identifying the needs and features required to develop the project successfully. In this project, the requirements were gathered based on user needs and the objective of providing interactive visualization for electric vehicle data.

3.1 Customer Journey Map

The customer journey map represents the steps a user follows while interacting with the visualization tool. It begins with exploring the dashboard, applying filters, comparing electric vehicles, analyzing charging stations, and finally gaining useful insights for decision-making. This helped in designing a userfriendly and interactive dashboard.



3.2 Solution Requirement

The project requires EV datasets containing information about vehicle specifications, charging stations, pricing, efficiency, and powertrain types. Tableau is used to create interactive dashboards and stories, while MySQL is used for storing and preprocessing the data. The solution provides users with meaningful visual insights through easy-to-use dashboards.

Functional Requirements:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Import	Import EV datasets (CSV files) into MySQL and connect them to Tableau. Validate imported data before visualization.
FR-2	Dashboard Visualization	Display interactive dashboard with EV analysis. Show charts for Top Speed, Body Style, Efficiency, Brands, and Powertrain Type.
FR-3	Story Presentation	Display Tableau Story with step-by-step insights. Include navigation between Dashboard and Story.
FR-4	Data Filtering	Allow users to filter data by Brand, Body Style, Powertrain Type, Region, and Charging Station Type. Update charts dynamically based on selected filters.
FR-5	Web Application	Embed Tableau Dashboard and Story into Bootstrap website. Provide responsive interface for users.

FR-6	Data Analysis	Compare EV brands, charging stations, efficiency, prices, and top speed. Present insights through interactive visualizations.

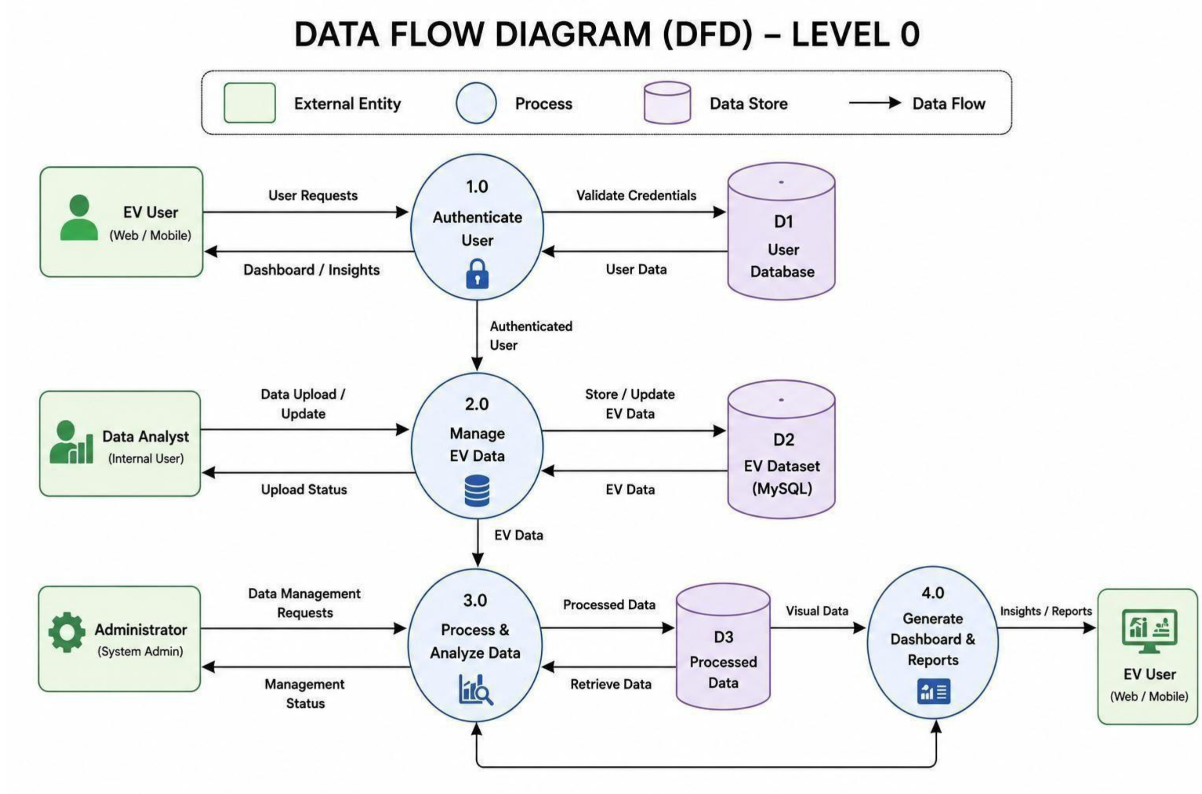
Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR1	**Usability**	The application should be easy to navigate with a simple and user-friendly interface.
NFR2	**Security**	The dashboard should use published Tableau Public links without modifying the original data.
NFR3	**Reliability**	The dashboard should display accurate and consistent visualizations based on the uploaded datasets.
NFR4	**Performance**	Dashboard and Story should load quickly and respond smoothly to filters.

3.3 Data Flow Diagram

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.



User Stories

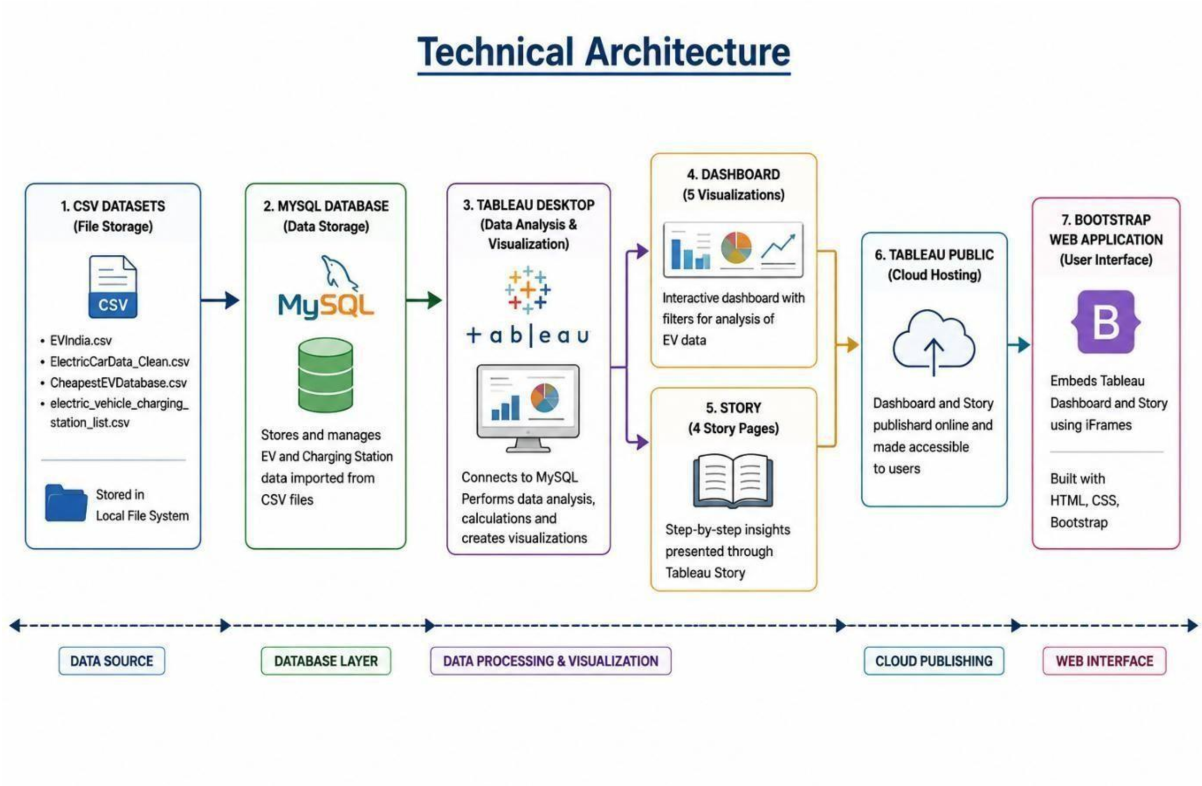
Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
EV User	Dashboard	USN-1	As a user, I can open the EV dashboard and view overall EV	Dashboard displays all charts and KPIs successfully.	High	Sprint-1
EV User	Vehicle Information	USN-2	As a user, I can view different EV brands, prices, top speed, and efficiency.	Vehicle information is displayed correctly.	High	Sprint-1

EV User	Apply Filters	USN-3	As a user, I can filter the dashboard by brand, body style, and powertrain type.	All visualizations update based on selected filters.	High	Sprint-1
EV User	Charging Station Map	USN-4	As a user, I can view charging stations across different regions of India	Charging station map loads correctly with regional data.	Medium	Sprint-2
EV User	Dashboard Insights	USN-5	As a user, I can compare EV performance using interactive dashboards.	Dashboard provides meaningful comparisons and insights.	High	Sprint-2
Data Analyst	Dataset Management	USN-6	As an analyst, I can update EV datasets whenever new information is available.	Updated data is reflected in Tableau after refresh.	Medium	Sprint-2
Administrator	Data Maintenance	USN-7	As an administrator, I can maintain and validate the EV datasets stored in MySQL.	Database remains updated and error-free.	Low	Sprint-2

3.4 Technology Stack

The project uses Tableau for data visualization, MySQL for data storage and preprocessing, and CSV files as the primary data source. The dashboards and stories created in Tableau provide an interactive platform for analyzing electric vehicle charge and range data.



1.	User Interface	Web interface for viewing dashboard and story	HTML, CSS, Bootstrap
2.	Application Logic-1	Data filtering, sorting and visualization	Tableau Desktop
3.	Application Logic-2	Interactive Dashboard and Story	Tableau Dashboard & Story
4.	Database	Stores EV datasets	MySQL

4. PROJECT DESIGN

The project design phase focuses on developing a solution that effectively analyzes and visualizes electric vehicle data. The system is designed to provide interactive dashboards and stories that help users explore EV information through simple and meaningful visualizations.

4.1 Problem Solution Fit

The proposed solution addresses the challenge of analyzing large EV datasets by converting them into interactive visualizations. It enables users to compare vehicle specifications, charging infrastructure, and performance, making EV data easier to understand and analyze.

Element	Details
Target Customer	EV buyers and policy analysts in India and globally
Observed Problem	Difficulty in comparing EV cars across brands and features; lack of centralized visual data on charging infrastructure
Why It's a Problem	Users must manually browse multiple sources and spreadsheets, causing confusion, time delays, and uncertainty in decisionmaking
Your Solution	A comprehensive Tableau dashboard showing EV performance, brand comparisons, price, speed, efficiency, and charging station mapping

Purpose

- Develop solutions that effectively address complex customer challenges based on their current needs and circumstances.
- Improve the speed of market adoption by aligning your solution with customers' existing habits, platforms, and preferred channels.
- Strengthen marketing and communication efforts by delivering the right message at the right time using relevant customer triggers.
- Build stronger customer relationships by solving recurring frustrations, urgent issues, or highimpact problems, while increasing meaningful interactions with your organization.
- Analyze the current customer experience to identify opportunities for improvement and deliver greater value to the target audience.

Why It Works	Centralizes insights using clear visuals, filters, and story dashboards — making comparisons quicker and easier for both consumers and planners

How You Validate Fit	Tested by students and reviewers; dashboards clarified topperforming EVs, charging gaps, and brand-wise models, proving useful for real decision-making

4.2 Proposed Solution

The proposed solution is an interactive Tableau dashboard that presents electric vehicle data using charts, maps, and graphs. Users can apply filters to compare EV brands, charging stations, body styles, powertrain types, pricing, and efficiency. The Tableau Story further explains the key insights in a step-bystep manner.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Many prospective electric vehicle (EV) buyers and policymakers face difficulties in obtaining clear, organized, and visually informative data about EV models and charging infrastructure. Since there is no single platform that brings this information together for easy comparison, users often experience confusion, making the decision-making process slower and less efficient.
2.	Idea / Solution description	The proposed solution is an interactive dashboard developed using Tableau, supported by datasets containing information on electric vehicle models and charging stations in India as well as other countries. The dashboard enables users to analyze EV specifications, compare different brands, examine pricing and top speed, explore charging station availability, and gain regionspecific insights through interactive charts, visualizations, and filtering options.

3.	Novelty / Uniqueness	While many sites provide EV specs, our solution is unique in integrating multiple datasets (India + global), using a visual-first approach. It combines EV model insights, charging data, and user-friendly filtering in a single interactive platform.
4.	Social Impact / Customer Satisfaction	Our dashboard enables eco-conscious consumers to make informed EV purchase decisions. It also helps policymakers and analysts plan infrastructure more effectively. This contributes to smarter mobility and supports India's EV adoption push.

The proposed solution is an interactive Tableau dashboard that presents electric vehicle data using charts, maps, and graphs. Users can apply filters to compare EV brands, charging stations, body styles, powertrain types, pricing, and efficiency. The Tableau Story further explains the key insights in a step-bystep manner.

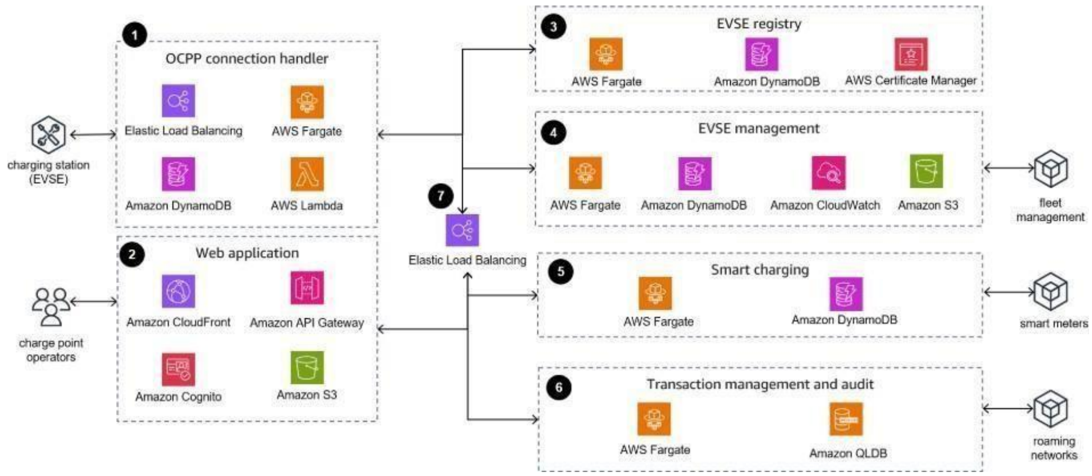
4.3 Solution Architecture

Solution architecture is a structured approach that connects business requirements with appropriate technological solutions. It involves designing a system that effectively addresses organizational challenges while ensuring that the proposed solution aligns with business objectives and technical requirements.

Objectives

- Identify the most suitable technology-based solution to address existing business challenges.
- Present the software's overall structure, functionality, behavior, and key components to stakeholders in a clear and understandable manner.
- Outline the system's features, implementation stages, and functional as well as nonfunctional requirements.

Establish the technical guidelines and specifications needed for the successful design, development, deployment, and maintenance of the solution.



5. PROJECT PLANNING & SCHEDULING

Project planning and scheduling helped the team complete the project in a systematic and organized manner. The work was divided into different phases such as data collection, data preprocessing, dashboard development, story creation, testing, and documentation. Each phase was completed according to the planned timeline to ensure successful project completion.

5.1 Project Planning

The project was planned by assigning tasks to team members based on their responsibilities. Initially, EV datasets were collected and preprocessed using MySQL. After data preparation, interactive dashboards and stories were developed in Tableau. Finally, the project was tested, analyzed, and documented before submission. Proper planning and coordination among team members ensured the timely completion of the project

Sprint	Functional Requirement (Epic)	User Story No.	User Story / Task	Story Points	Priority	Team Members
Sprint1	Data Connection	USN-1	As a team, we can import all EV datasets into MySQL and verify schema	2	High	Chendilikumar kavya

Sprint1	Data Visualization (Cars)	USN-2	As a user, I can view EV car specs, speed, and price in comparative views	3	High	Vyshnavi
Sprint1	Data Visualization (Charging)	USN-3	As a user, I can view charging stations by region and type	2	High	Venkata Thimma Reddy
Sprint2	Filtering Features	USN-4	As a user, I can filter cars by powertrain, body style, or brand	2	Medium	Chamarthi Ajay Kumar Raju
Sprint2	Summary Cards	USN-5	As a user, I can see summary cards comparing Indian and global EV brands	2	Medium	Vyshnavi
Sprint2	Dashboard & Story	USN-6	As a user, I can view all sheets together in a dashboard & story layout	3	High	Chamarthi Ajay Kumar Raju
Sprint2	Publish Dashboard	USN-7	As a team, we can publish the dashboard to Tableau Public	1	High	Chendilikumar kavya

.

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	7	5 Days	7 July 2026	12 July 2026	7	14 July 2026
Sprint-2	8	6 Days	8 July 2026	14 July 2026	TBD	TBD

Velocity:

For Sprint-1:

Story Points Completed: **7**

Duration: **5 Days**

➞ Velocity = 7 ÷ 5 = 1.4 story points/day

Use of Velocity:

- To estimate future sprints more accurately.
- Based on this, Sprint-2 (8 points) would ideally need:

8 ÷ 1.4 ≈ 5.7 days → Round to 6 Days

6. FUNCTIONAL AND PERFORMANCE TESTING

Functional and performance testing was carried out to ensure that the visualization tool works correctly and provides accurate results. The dashboards, filters, calculations, and stories were tested to verify that users can interact with the visualizations without any issues. The application performed successfully and displayed the expected insights.

6.1 Performance Testing

Performance testing was conducted to evaluate the responsiveness and accuracy of the dashboards. The EV datasets were loaded successfully, filters worked correctly, and all visualizations were displayed without errors. The Tableau dashboards and stories responded quickly to user interactions, providing smooth navigation and effective data analysis.

The performance of the visualization tool was evaluated using multiple EV datasets containing vehicle specifications and charging station information. The dashboards loaded the data efficiently, and all filters, maps, charts, and calculated fields functioned as expected. Users were able to switch between different visualizations and explore the data smoothly without affecting the overall performance of the application.

Model Performance Testing:

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	Successfully imported and rendered **4 CSV datasets** into Tableau for visualization. **Datasets Used:** • **Cheapestelectriccars-EVDatabase.csv** – Contains information about affordable electric vehicles, including price and specifications. • **electric_vehicle_charging_station_list.csv** – Contains EV charging station details across different regions in India. • **ElectricCarData_Clean.csv** – Contains EV specifications such as range, battery capacity, efficiency, top speed, and powertrain information. • **EVIndia.csv** – Contains electric vehicle brand, model, body style, and related information used for comparison and analysis.
2.	Data Preprocessing	Cleaned the datasets by verifying data types, removing unnecessary columns, checking missing values, and preparing the data for visualization.
3.	Utilization of Filters	Filters used include **Brand**, **Body Style**, and **Powertrain Type** to allow interactive analysis of EV data.
4.	Calculation fields Used	Created a calculated field: Price / Efficiency Ratio to compare the cost of electric vehicles with their efficiency (km/kWh).

5.	Dashboard design	No of Visualizations / Graphs – 5 • Brands by Body Style • Top 10 Most Efficient EV Brands • Brands by Powertrain Type • Different EV Cars in India • Top Speed by Brand
6	Story Design	No of Visualizations / Graphs -4 • EV Charging Stations in India • Charging Stations by Region and Type • Price of Electric Cars by Different Brands • Different Brands and Number of Models

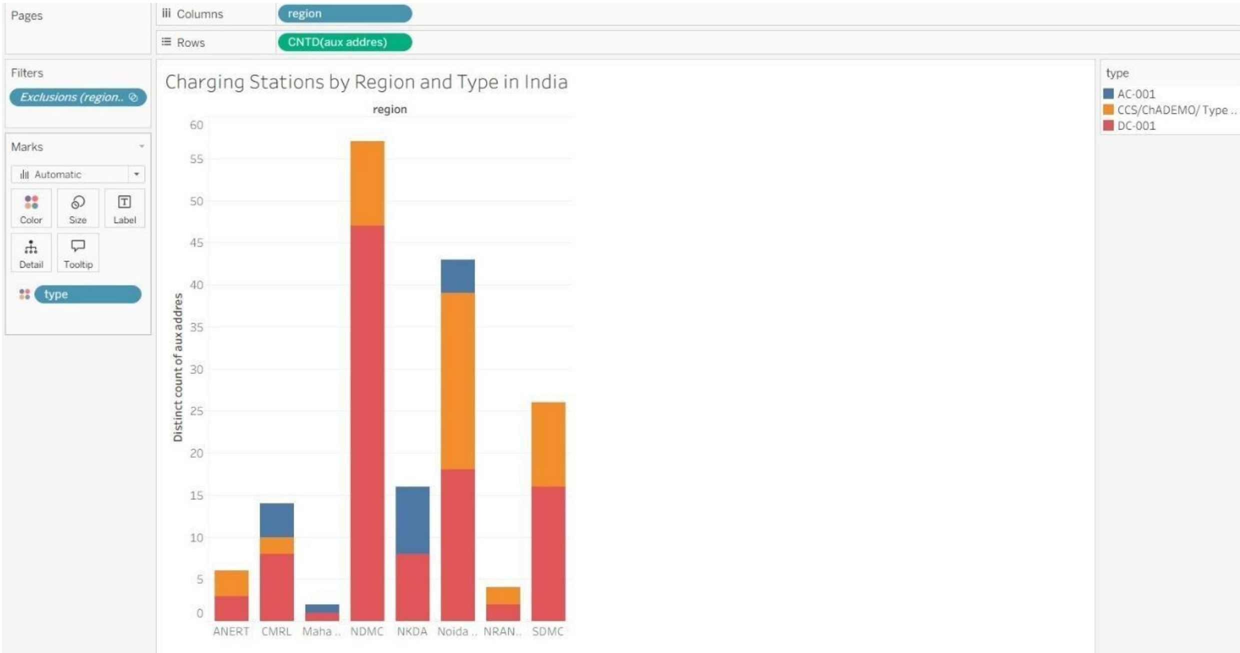
7. **RESULTS**

The project was successfully completed by developing interactive Tableau dashboards and stories for Electric Vehicle Charge and Range Analysis. The visualizations provide meaningful insights into charging infrastructure, vehicle specifications, pricing, efficiency, and brand comparisons. The results help users understand EV data in a clear and interactive manner, making analysis easier and supporting better decision-making.

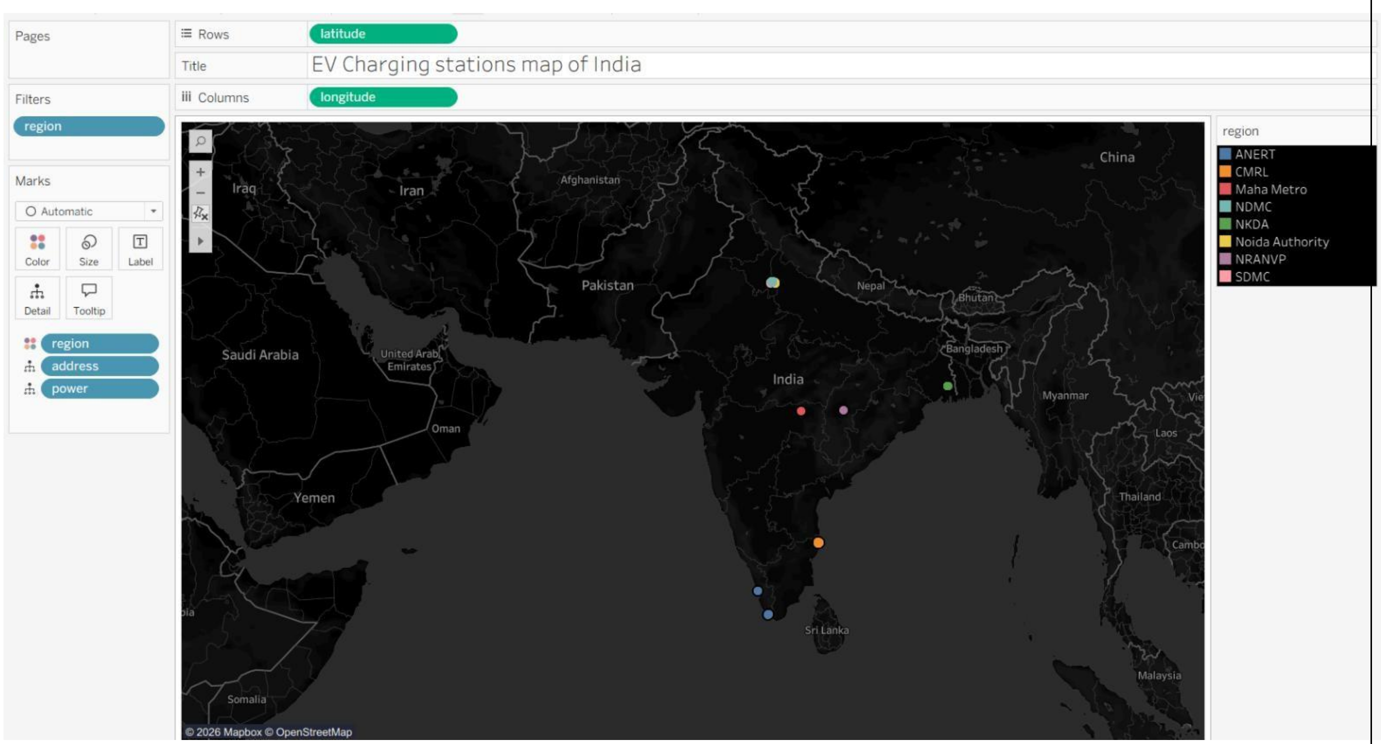
7.1 Output Screenshots

The output of the project is presented through interactive Tableau dashboards and stories. The dashboard includes visualizations such as Brands by Body Style, Top 10 Efficient EV Brands, Brands by Powertrain Type, Different EV Cars in India, and Top Speed by Brand. The Tableau Story further explains key insights using visualizations like Charging Stations in India, Charging Stations by Region and Type, Price of Electric Cars, and Different Brands and Number of Models. These visualizations provide a comprehensive view of the electric vehicle ecosystem and help users explore the data effectively.

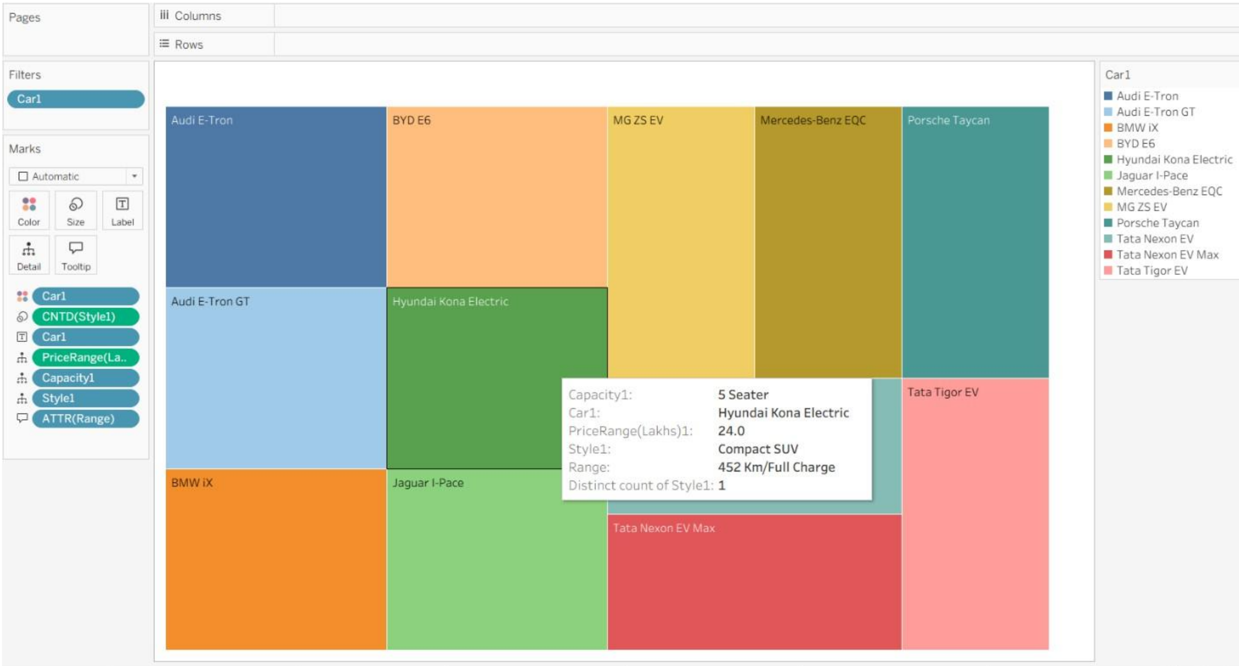
Visualizations:

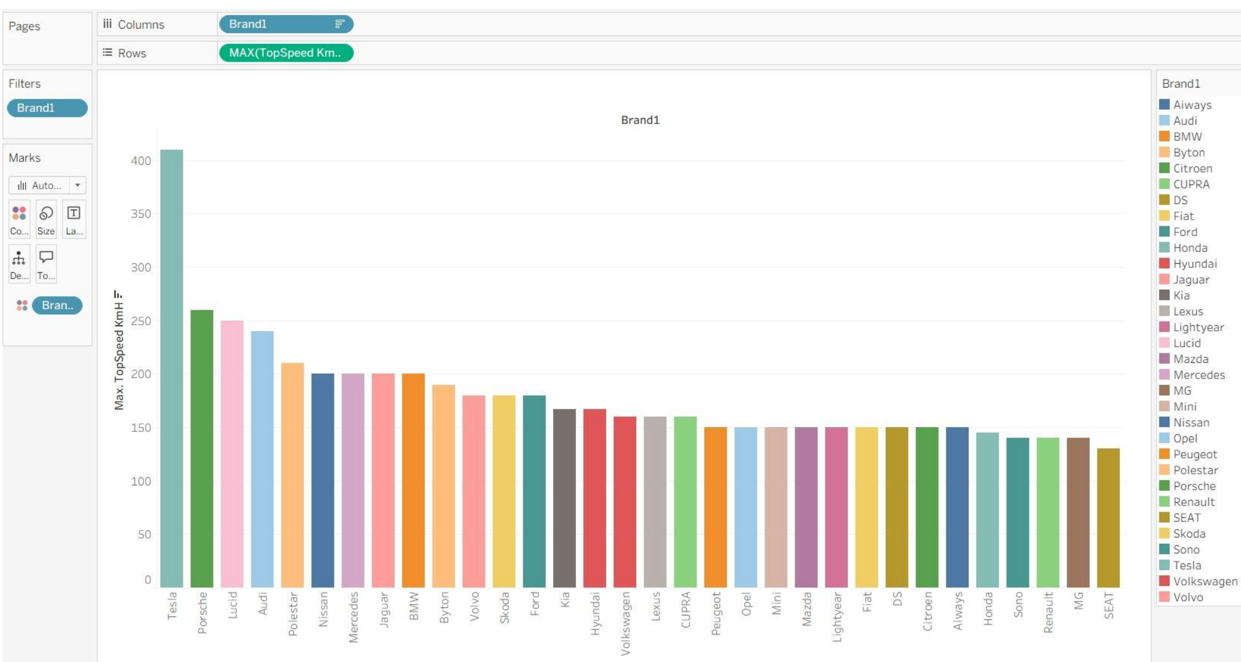


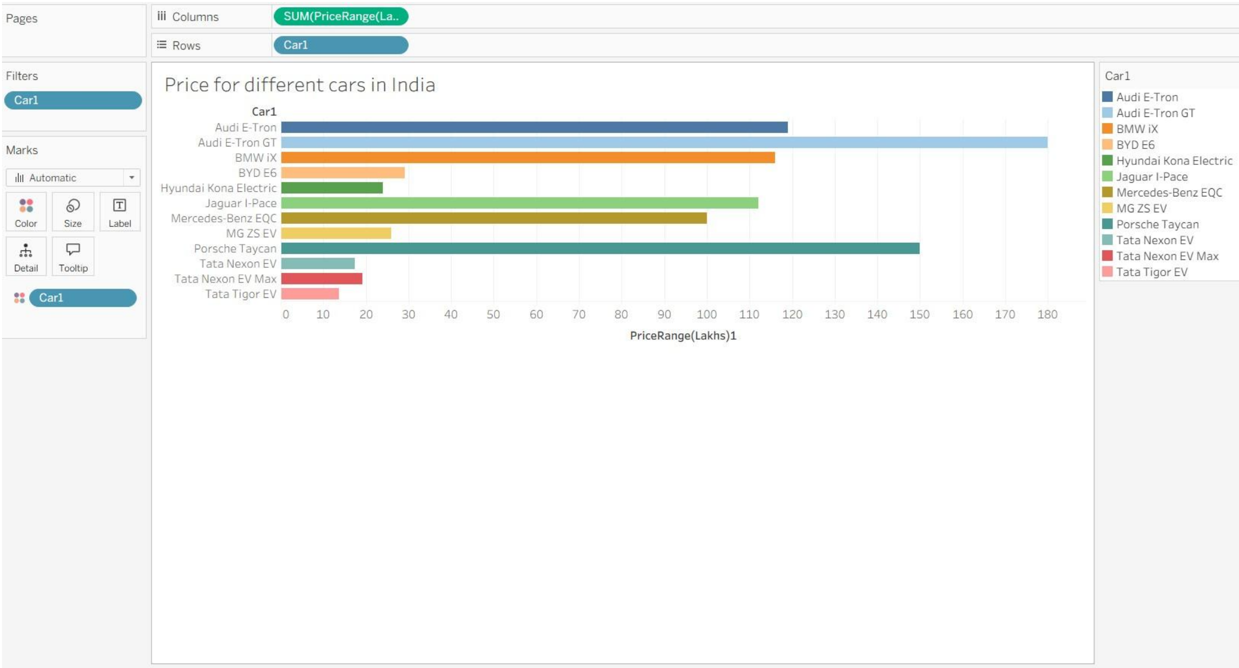
Pages
Columns
region
Rows
CNTD(aux addres)
Filters
Exclusions (region..
Marks
Automatic
Color
Size
Label
Detail
Tooltip
type
Charging Stations by Region and Type in India
region
Distinct count of aux addres
60
55
50
45
40
35
30
25
20
15
10
5
0
ANERT
CMRL
Maha ..
NDMC
NKDA
Noida ..
NRAN..
SDMC
type
AC-001
CCS/ChADEMO/ Type ..
DC-001

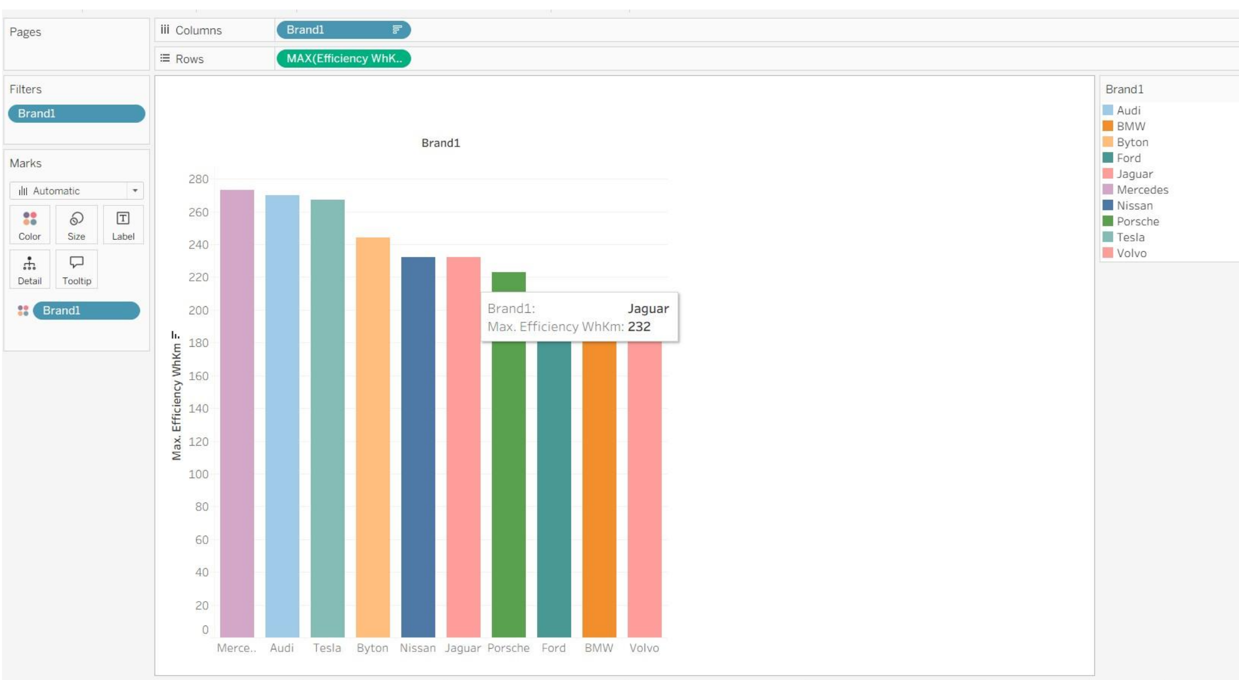


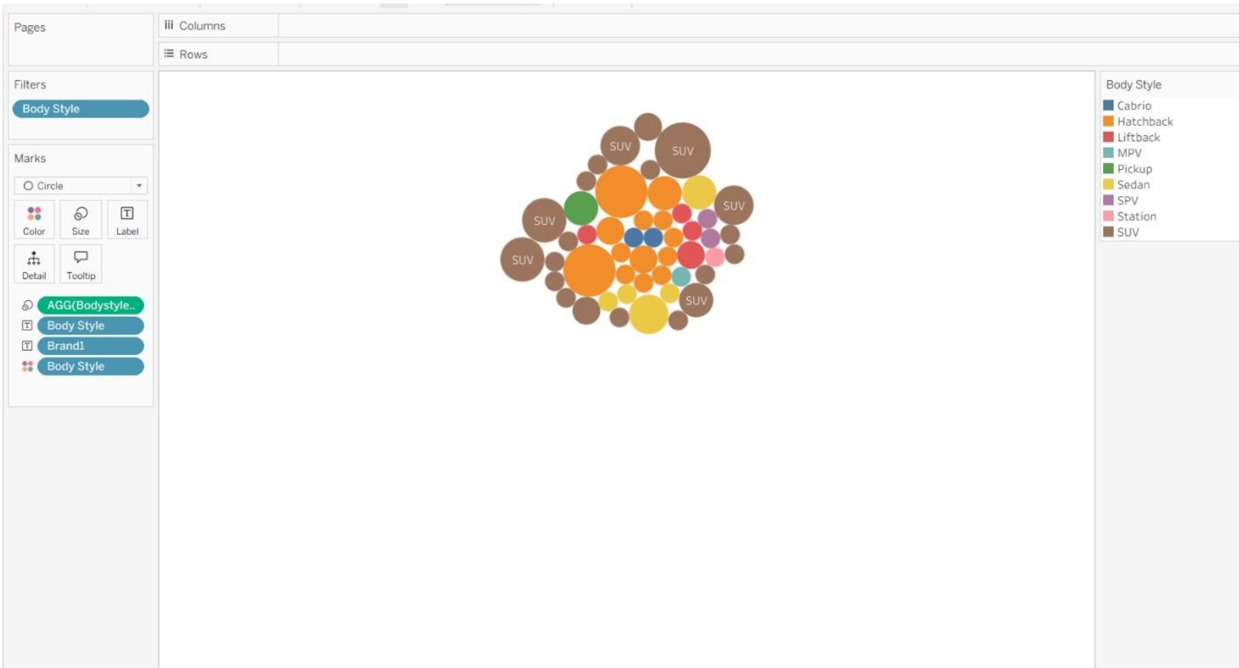
Pages
Rows
latitude
Title
EV Charging stations map of India
Columns
longitude
Filters
region
Marks
Automatic
Color
Size
Label
Detail
Tooltip
region
address
power
region
ANERT
CMRL
Maha Metro
NDMC
NKDA
Noida Authority
NRANVP
SDMC
China
Iraq
Iran
Afghanistan
Pakistan
Nepal
Bhutan
Bangladesh
Saudi Arabia
United Arab Emirates
India
Myanmar
Oman
Laos
Yemen
Thailand
Sri Lanka
Malaysia
Somalia
© 2026 Mapbox © OpenStreetMap

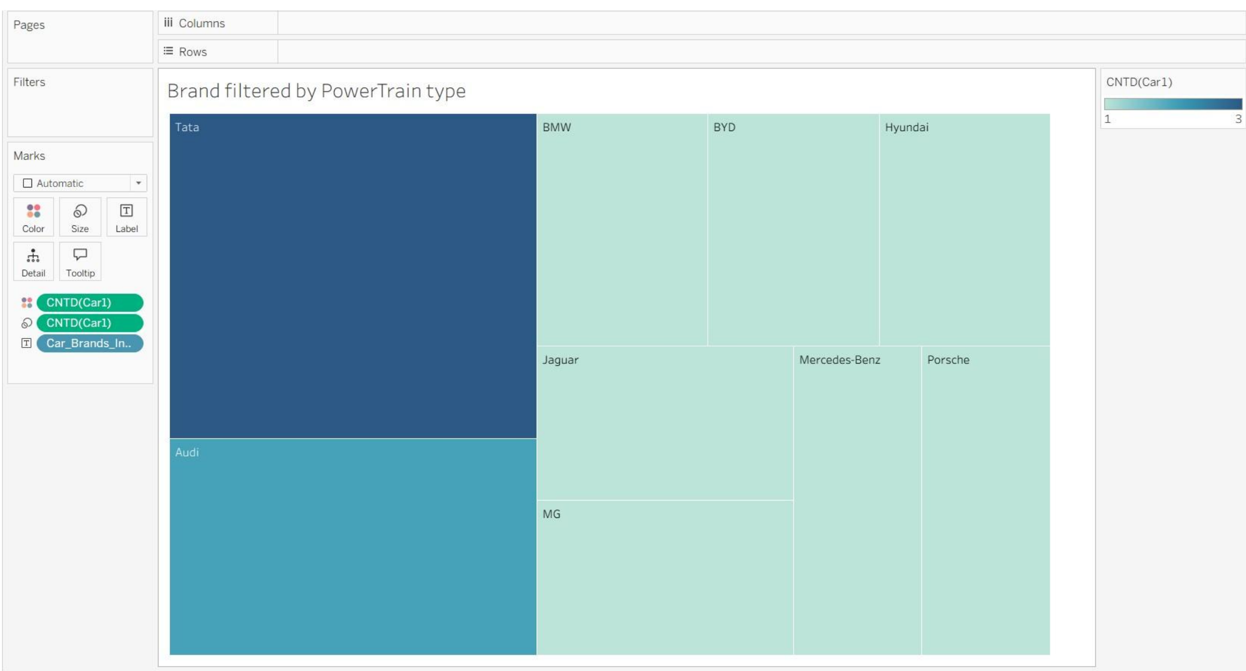

Pages
Columns
Rows
Filters
Car1
Marks
Automatic
Color
Size
Label
Detail
Tooltip
Car1
CNTD(Style1)
Car1
PriceRange(La..
Capacity1
Style1
ATTR(Range)
Audi E-Tron
BYD E6
MG ZS EV
Mercedes-Benz EQC
Porsche Taycan
Audi E-Tron GT
Hyundai Kona Electric
Tata Tigor EV
BMW iX
Jaguar I-Pace
Tata Nexon EV Max
Capacity1: 5 Seater
Car1: Hyundai Kona Electric
PriceRange(Lakhs)1: 24.0
Style1: Compact SUV
Range: 452 Km/Full Charge
Distinct count of Style1: 1
Car1
Audi E-Tron
Audi E-Tron GT
BMW iX
BYD E6
Hyundai Kona Electric
Jaguar I-Pace
Mercedes-Benz EQC
MG ZS EV
Porsche Taycan
Tata Nexon EV
Tata Nexon EV Max
Tata Tigor EV

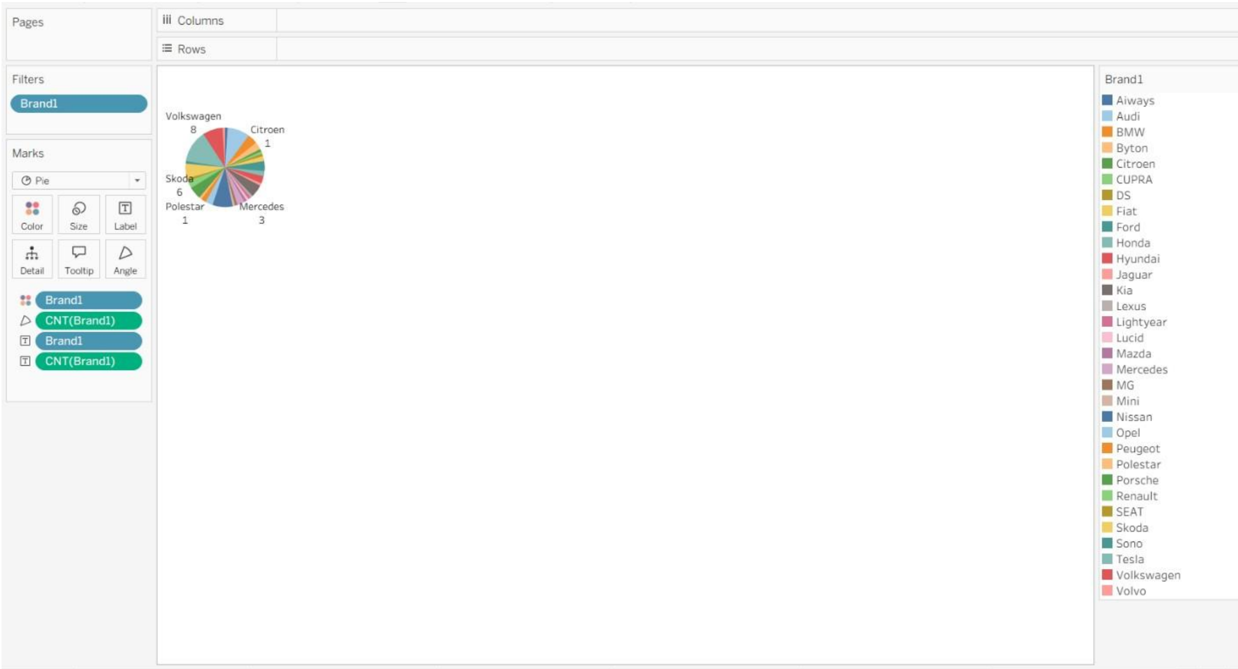

Pages
Columns
Brand1
Rows
MAX(TopSpeed Km..
Filters
Brand1
Marks
Auto...
Co...
Size
La...
De...
To...
Bran..
Brand1
Max. TopSpeed KmH
400
350
300
250
200
150
100
50
0
Tesla
Porsche
Lucid
Audi
Polestar
Nissan
Mercedes
Jaguar
BMW
Byton
Volvo
Skoda
Ford
Kia
Hyundai
Volkswagen
Lexus
CUPRA
Peugeot
Opel
Mini
Mazda
Lightyear
Fiat
DS
Citroen
Aiways
Honda
Sono
Renault
MG
SEAT
Brand1
Aiways
Audi
BMW
Byton
Citroen
CUPRA
DS
Fiat
Ford
Honda
Hyundai
Jaguar
Kia
Lexus
Lightyear
Lucid
Mazda
Mercedes
MG
Mini
Nissan
Opel
Peugeot
Polestar
Porsche
Renault
SEAT
Skoda
Sono
Tesla
Volkswagen
Volvo


Pages
Columns
SUM(PriceRange(La..
Rows
Car1
Filters
Car1
Marks
Automatic
Color
Size
Label
Detail
Tooltip
Car1
Price for different cars in India
Car1
Audi E-Tron
Audi E-Tron GT
BMW iX
BYD E6
Hyundai Kona Electric
Jaguar I-Pace
Mercedes-Benz EQC
MG ZS EV
Porsche Taycan
Tata Nexon EV
Tata Nexon EV Max
Tata Tigor EV
0 10 20 30 40 50 60 70 80 90 100 110 120 130 140 150 160 170 180
PriceRange(Lakhs)1

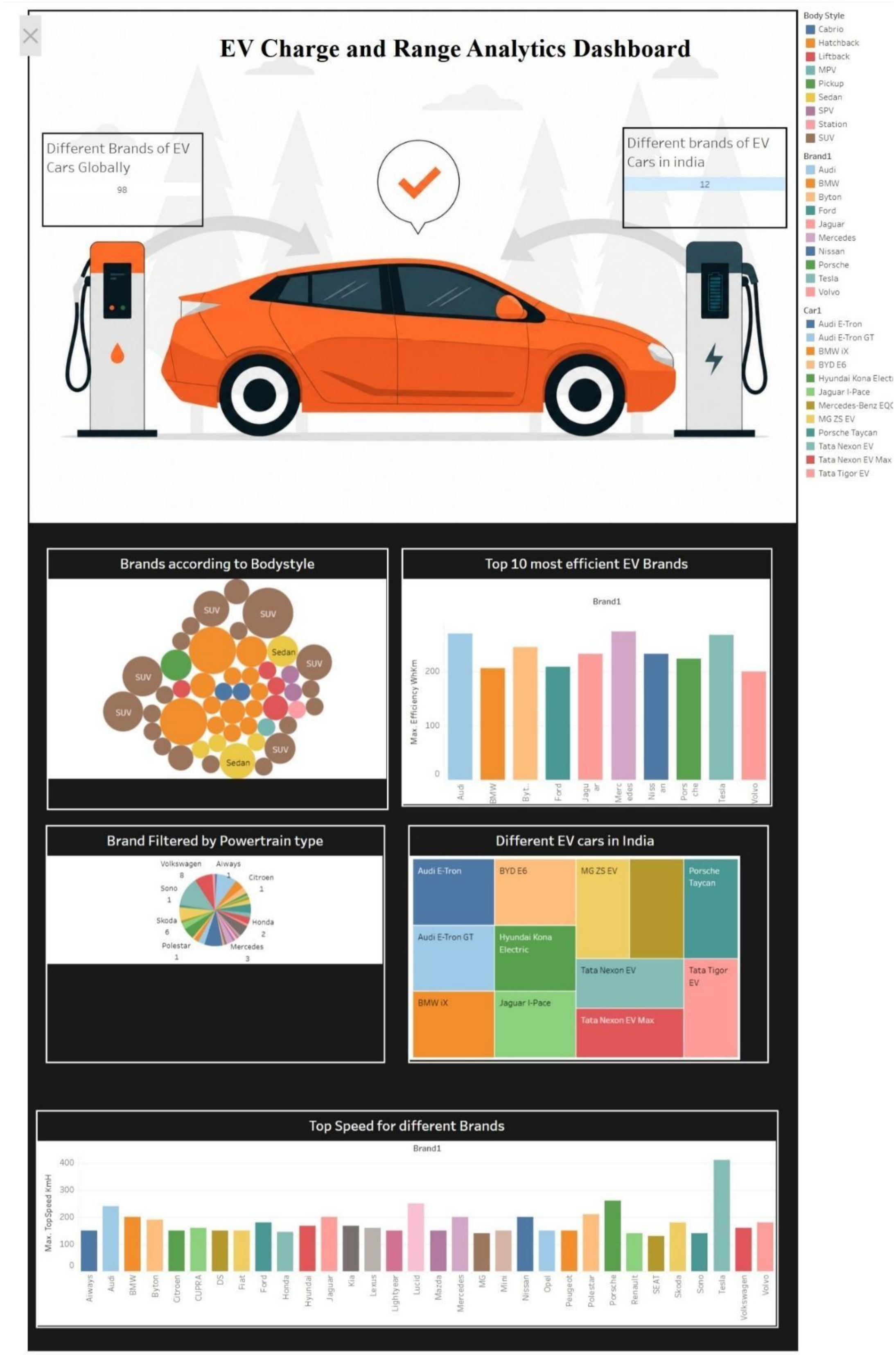

Pages
Columns
Brand1
Rows
MAX(Efficiency WhK..
Filters
Brand1
Marks
Automatic
Color
Size
Label
Detail
Tooltip
Brand1
Brand1
Audi
BMW
Byton
Ford
Jaguar
Mercedes
Nissan
Porsche
Tesla
Volvo
Max. Efficiency WhKm
0 20 40 60 80 100 120 140 160 180 200 220 240 260 280
Brand1: Jaguar
Max. Efficiency WhKm: 232
Merce.. Audi Tesla Byton Nissan Jaguar Porsche Ford BMW Volvo


Pages
Columns
Rows
Filters
Body Style
Marks
Circle
Color
Size
Label
Detail
Tooltip
AGG(Bodystyle..
Body Style
Brand1
Body Style
Body Style
Cabrio
Hatchback
Liftback
MPV
Pickup
Sedan
SPV
Station
SUV

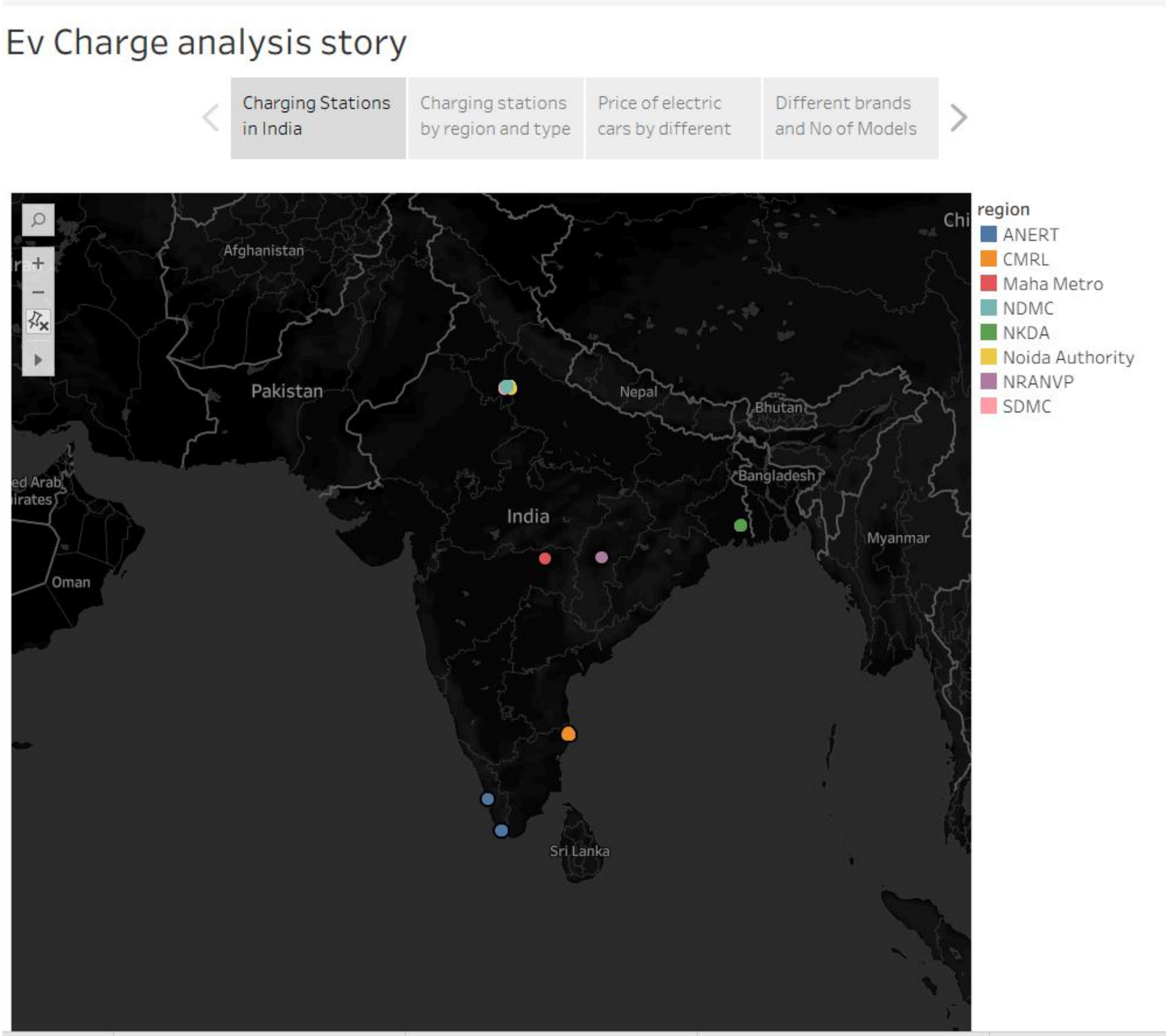

Pages
Columns
Rows
Filters
Marks
Automatic
Color
Size
Label
Detail
Tooltip
CNTD(Car1)
CNTD(Car1)
Car_Brands_In..
Brand filtered by PowerTrain type
CNTD(Car1)
1
3
Tata
BMW
BYD
Hyundai
Jaguar
Mercedes-Benz
Porsche
Audi
MG


Pages
Columns
Rows
Filters
Brand1
Marks
Pie
Color
Size
Label
Detail
Tooltip
Angle
Brand1
CNT(Brand1)
Brand1
CNT(Brand1)
Volkswagen
8
Citroen
1
Skoda
6
Polestar
1
Mercedes
3
Brand1
Aiways
Audi
BMW
Byton
Citroen
CUPRA
DS
Fiat
Ford
Honda
Hyundai
Jaguar
Kia
Lexus
Lightyear
Lucid
Mazda
Mercedes
MG
Mini
Nissan
Opel
Peugeot
Polestar
Porsche
Renault
SEAT
Skoda
Sono
Tesla
Volkswagen
Volvo

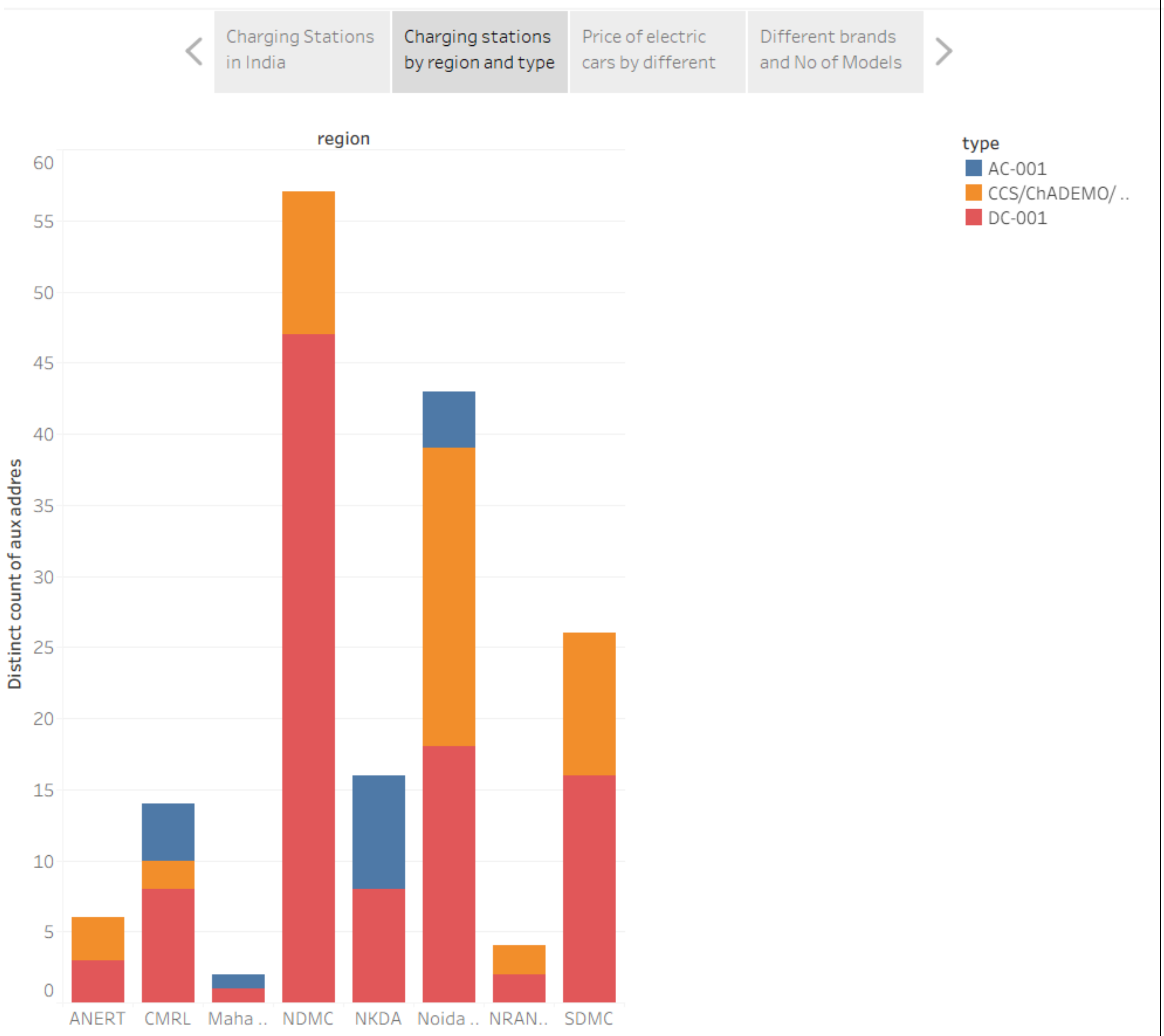
Dashboard:

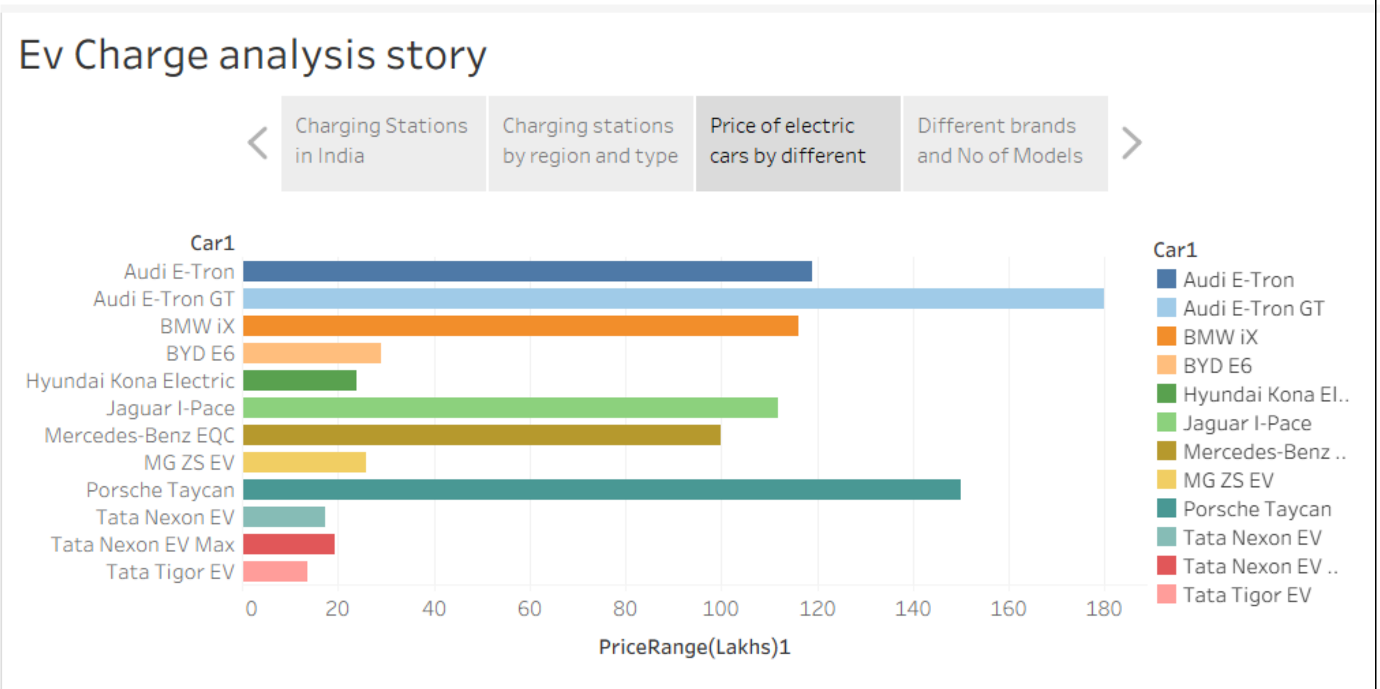

EV Charge and Range Analytics Dashboard
Different Brands of EV Cars Globally
98
Different brands of EV Cars in india
12
Body Style
Cabrio
Hatchback
Liftback
MPV
Pickup
Sedan
SPV
Station
SUV
Brand1
Audi
BMW
Byton
Ford
Jaguar
Mercedes
Nissan
Porsche
Tesla
Volvo
Car1
Audi E-Tron
Audi E-Tron GT
BMW iX
BYD E6
Jaguar I-Pace
MG ZS EV
Porsche Taycan
Tata Nexon EV
Tata Nexon EV Max
Tata Tigor EV
Brands according to Bodystyle
Top 10 most efficient EV Brands
Brand Filtered by Powertrain type
Different EV cars in India
Audi E-Tron
BYD E6
MG ZS EV
Porsche Taycan
Audi E-Tron GT
Hyundai Kona Electric
Tata Nexon EV
Tata Tigor EV
BMW iX
Jaguar I-Pace
Tata Nexon EV Max
Top Speed for different Brands
Brand1
Max. Top Speed KmH

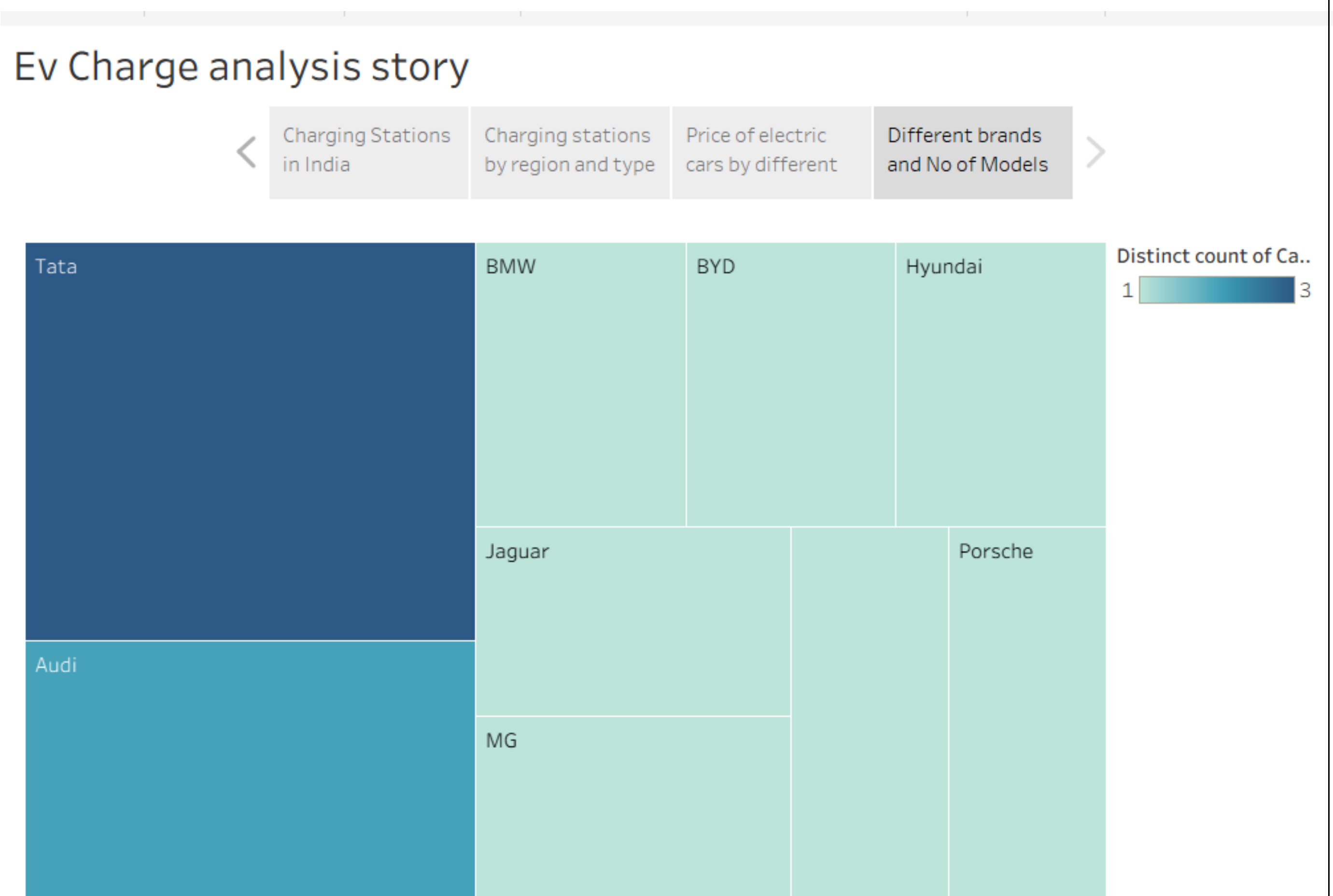
Story:



Ev Charge analysis story
Charging Stations in India
Charging stations by region and type
Price of electric cars by different
Different brands and No of Models
Afghanistan
Pakistan
Nepal
Bhutan
Bangladesh
India
Myanmar
Oman
Sri Lanka
region
ANERT
CMRL
Maha Metro
NDMC
NKDA
Noida Authority
NRANVP
SDMC


Charging Stations in India
Charging stations by region and type
Price of electric cars by different
Different brands and No of Models
region
type
AC-001
CCS/ChADEMO/ ..
DC-001
Distinct count of aux addres
60
55
50
45
40
35
30
25
20
15
10
5
0
ANERT
CMRL
Maha ..
NDMC
NKDA
Noida ..
NRAN..
SDMC


Ev Charge analysis story
Charging Stations in India
Charging stations by region and type
Price of electric cars by different
Different brands and No of Models
Car1
Audi E-Tron
Audi E-Tron GT
BMW iX
BYD E6
Hyundai Kona Electric
Jaguar I-Pace
Mercedes-Benz EQC
MG ZS EV
Porsche Taycan
Tata Nexon EV
Tata Nexon EV Max
Tata Tigor EV
0 20 40 60 80 100 120 140 160 180
PriceRange(Lakhs)1
Car1
Audi E-Tron
Audi E-Tron GT
BMW iX
BYD E6
Hyundai Kona El..
Jaguar I-Pace
Mercedes-Benz ..
MG ZS EV
Porsche Taycan
Tata Nexon EV
Tata Nexon EV ..
Tata Tigor EV


Ev Charge analysis story
Charging Stations in India
Charging stations by region and type
Price of electric cars by different
Different brands and No of Models
Tata
BMW
BYD
Hyundai
Audi
Jaguar
Porsche
MG
Distinct count of Ca..
1
3

8. **ADVANTAGES & DISADVANTAGES**

Advantages :
1.Provides an interactive dashboard for easy visualization of EV data.
2.Helps users compare different EV brands, models, charging stations, and price ranges.
3.Filters enable users to analyze data based on brand, body style, and powertrain type.
4.Tableau dashboards and stories present information in a simple and attractive manner.
5.Supports better decision-making for customers interested in electric vehicles.

Disadvantages :
1.The analysis depends on the available dataset and may not include newly launched EV models.
2.Real-time EV charging station information is not available.
3.Internet access is required to view the Tableau Public dashboard.
4.The project is limited to the selected dataset and does not provide live market updates.
5.Future enhancements can improve accuracy by integrating real-time data sources.

9. **CONCLUSION**

The EV Charge and Range Analysis project was successfully developed using Tableau to provide an interactive dashboard and story for analyzing electric vehicle data. The project enables users to explore EV brands, charging stations, price ranges, efficiency, and vehicle specifications through clear visualizations. The dashboard and story make complex data easy to understand and support better decision-making. Overall, the project demonstrates how data visualization can help users gain meaningful insights into the growing electric vehicle ecosystem.

10.FUTURE SCOPE

The project can be enhanced by integrating real-time electric vehicle data and live charging station information. Future versions can include EV recommendations based on user preferences, route planning with nearby charging stations, and charging time estimation. Additional features such as predictive analytics, battery health analysis, and mobile application support can further improve the usability of the system. These enhancements will make the project more intelligent, accurate, and useful for EV users and researchers.

11. **APPENDIX:**

Dataset Link:
https://drive.google.com/drive/folders/1Rkzdks6Us1Uq2SRB4nxMAb83jN5bpHll

Tableau Public Link:

Dashboard:
https://public.tableau.com/app/profile/kavya.chendilikumar/viz/EV_Charge_Range_Analysiss/EVchargeandrangeanalysisdashboard

Story:
https://public.tableau.com/shared/DMWHYY49Y?:display_count=n&:origin=viz_share_link

GitHub Repository:
https://github.com/ckavya56/visualization-tool-for-electric-vehicle-charge-and-range-analysis

Project Demo link:
https://drive.google.com/file/d/1HGv_Slw0lyn4qcjW5cRF16vWaiKnYh-w/view?usp=sharing